AMIT SHIVAM

Postdoctoral Researcher, Guidance, Control and Estimation for Autonomous Vehicles | Porto, Portugal

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Summary

Postdoctoral researcher in guidance, control and estimation for fixed-wing and multirotor UAVs. I design guidance laws, safety filters and estimators with formal guarantees (convergence, bounded curvature, feasibility under actuator saturation) and validate them through simulation, software-in-the-loop and hardware-in-the-loop testing to outdoor flight trials and real-time hardware. Ph.D., Indian Institute of Science (2024). Seventeen publications, including three journal articles (two in the *Journal of Guidance, Control, and Dynamics*, one in *Aerospace Science and Technology*). MSCA Seal of Excellence (90%). Finalist (1 of 6), TrustRobotics Challenge, CSAW'26 Europe. Postdoctoral position at the University of Porto and Politecnico di Milano, with supervision of one Ph.D., two M.Sc. and three B.Sc. projects.

Core Expertise

Guidance	Vector-field path following for fixed-wing and multirotor UAVs with convergence and bounded-curvature guarantees: Arcsine, Lamé-curve, variable look-ahead, geometric look-angle shaping (GLASS), robust and multi-UAV vector fields.
Trajectory generation	Curvature-continuous, dynamically feasible paths for surveillance and inspection (Lamé-curve families); airborne wind energy trajectories under tether constraints.
Robust control	Hybrid-gain finite-time sliding-mode and prescribed-performance control; feasibility analysis under actuator saturation and bounded disturbances.
Safety-critical control	Lookahead-augmented control-barrier / control-Lyapunov-function quadratic programs for collision avoidance and reference tracking, run in real time on cars.
Estimation	Kalman and adaptive filtering; Koopman-based noise adaptation for time-varying, non-Gaussian uncertainty (in preparation); nonlinear target tracking.
Flight software and testing	Modified the ArduPilot C++ source to run variable look-ahead guidance in fixed-wing flight tests; ArduPilot SITL and HITL, ROS/ROS 2, Gazebo, QGroundControl; indoor Crazyflie trials with OptiTrack; outdoor multirotor and fixed-wing trials.
Tools	MATLAB/Simulink, Python, C++ (ArduPilot), ROS/ROS 2, Gazebo, LaTeX.

Experience

University of Porto, SYSTEC–ISR, ARISE Aug. 2025 – Present
Postdoctoral Researcher (supervisor: Prof. Fernando A. C. C. Fontes)

- Developed geometric look-angle shaping (GLASS) and variable look-ahead guidance that enlarge the operating envelope for path following; developed set-stabilising guidance for airborne wind energy cycles.
- Developed robust controllers with feasibility certificates under actuator saturation; developing a Koopman-based adaptive filter for non-Gaussian noise.
- Implemented variable look-ahead guidance inside the ArduPilot C++ source for fixed-wing flight tests; building ROS 2/Gazebo/ArduPilot SITL and HITL workflows for deployment-oriented validation.
- Supervise the student-led LACART team (CSAW'26 Europe finalist) and the Ph.D., M.Sc. and B.Sc. projects listed under Mentoring.

Politecnico di Milano, DEIB Aug. 2024 – Jul. 2025
Postdoctoral Researcher (supervisors: Prof. G. P. Incremona and Prof. L. Fagiano)

- Curvature-continuous Lamé-curve trajectory generation for boundary surveillance with guaranteed vehicle feasibility and reduced path length.
- Variable look-ahead and geometric guidance for airborne wind energy systems under tether and flight constraints.

Selected Projects

- **Variable L_0 guidance, flight tested.** Look-ahead distance that varies in time, widening the envelope in which a UAV converges onto a path. Implemented in ArduPilot, checked in SITL/HITL, flown on a hand-launched fixed-wing UAV with ground-station telemetry review (IFAC World Congress 2026; arXiv:2512.06124).
- **LACART safety filter, real time on hardware.** Lookahead CBF-CLF quadratic program for a unicycle on a 1:43-scale racing car with overhead vision, UDP state streaming and Bluetooth LE: 25 Hz loop, 7.49 ms

mean QP solve, 62.44 ms mean state age; reported limits are steering saturation, communication delay and track-boundary contact. Finalist, TrustRobotics Challenge CSAW'26 Europe.

- **Lamé-curve boundary surveillance.** Guidance law that circumscribes a rectangular boundary with a curvature-limited Lamé curve and converges from anywhere in the plane; MATLAB simulations and indoor Crazyflie validation (OptiTrack) (JGCD 2025; code).

Education

Ph.D., Aerospace Engineering — Indian Institute of Science, Bengaluru 2018 – 2024
Thesis: A Class of Vector Fields for Path Following Guidance (advisor: Prof. Ashwini Ratnoo). Awarded Dec. 2024

M.E., Control Systems Engineering — Jadavpur University, Kolkata 2015 – 2017
University Medal, First Rank (CGPA 9.22/10). Thesis: nonlinear estimation for 2D and 3D target tracking

Publications

Seventeen in total; see Google Scholar.

Journal articles

[J1] A. Shivam, K. Kumari, A. Ratnoo, "Robust Vector Field Guidance," *Aerospace Science and Technology*, 181(3), 2026. doi:10.1016/j.ast.2026.113855

[J2] A. Shivam, A. Ratnoo, "Surveillance Guidance Using Lamé Curve Paths," *J. Guidance, Control, and Dynamics*, 48(5):1124–1137, 2025. doi:10.2514/1.G008574

[J3] A. Shivam, A. Ratnoo, "Arcsine Vector Field for Path Following Guidance," *J. Guidance, Control, and Dynamics*, 46(12):2409–2420, 2023. doi:10.2514/1.G007562

Conference papers

[C1] A. Shivam, A. J. Gallo, G. P. Incremona, A. Ferrara, "Lamé Curve Path Generation for Robust Surveillance and Tracking," IFAC World Congress, 2026 (accepted).

[C2] A. Shivam, M. C. R. M. Fernandes, F. A. C. C. Fontes, L. Fagiano, "Variable L_0 Guidance Strategy: Enlarged Operational Envelope and Path-Following," IFAC World Congress, 2026 (accepted). arXiv:2512.06124

[C3] G. Kumar, A. Shivam, A. Ratnoo, "Multi-UAV Path Following using Vector-Field Guidance," Modeling, Estimation and Control Conf. (MECC), 2026 (accepted). arXiv:2604.17995

[C4] A. Shivam, N. K. Singh, L. T. Paiva, F. A. C. C. Fontes, "Direct Vector Field Guidance for Path Following," CONTROLO, 2026 (accepted). arXiv:2610.13576

[C5] A. Shivam, K. Kumari, F. A. C. C. Fontes, "Robust Control Design Using a Hybrid-Gain Finite-Time Sliding Mode Controller," European Control Conf. (ECC), 2026. IEEE Xplore

[C6] A. Shivam et al., "Feasibility Limits of Prescribed Performance Control Under Actuation Constraints," ECC, 2026 (accepted, late-breaking results).

[C7] A. Shivam, M. C. R. M. Fernandes, S. Vinha, F. A. C. C. Fontes, "Geometric Look-Angle Shaping Strategy for Enclosed Inspection," Int. Conf. Unmanned Aircraft Systems (ICUAS), 2026. doi:10.1109/ICUAS69441.2026.11598560

[C8] A. Shivam, M. C. R. M. Fernandes, S. Vinha, F. A. C. C. Fontes, "Set-Stabilizing Vector Field Guidance for Complete Airborne Wind Energy Cycles," Airborne Wind Energy Conf. (AWEC), Porto, 2026.

[C9] M. C. R. M. Fernandes, A. Shivam, F. A. C. C. Fontes, L. Fagiano, "Variable L_0 Path-Following Guidance for Airborne Wind Energy Systems," IFAC Workshop on LPV Systems, *IFAC-PapersOnLine* 59(16):128–133, 2025. doi:10.1016/j.ifacol.2025.10.091

[C10] A. Shivam, S. Neon, A. Ratnoo, "Bézier Curve Based Guidance for Autonomous Landing of Quadrotor," IEEE CoDIT, Rome, 2023, pp. 1308–1312. doi:10.1109/CoDIT58514.2023.10284179

[C11] A. Shivam, A. Ratnoo, "Vector Field for Following Elliptic Paths," AIAA SciTech Forum, 2022, Paper 2022-0538. doi:10.2514/6.2022-0538

[C12] A. Shivam, A. Ratnoo, "Curvature-Constrained Vector Field for Path Following Guidance," IEEE ICUAS, 2021, pp. 853–857. doi:10.1109/ICUAS51884.2021.9476741

Preprint and competition entry

[P1] A. Shivam, K. Kumari, F. A. C. C. Fontes, "Prescribed-Performance-Aware Hybrid-Gain-Based Robust Controller," under revision, 2026. arXiv:2602.14382

[P2] P. Lopes, A. Shivam, L. T. Paiva, "LACART: Lookahead-Augmented Collision Avoidance and Reference Tracking Strategies for Safety-Critical Robots," TrustRobotics Challenge, CSAW'26 Europe, Valence, 13 Nov. 2026 (extended abstract; one of 6 finalists).

Awards, Honours and Grants

- **MSCA Seal of Excellence**, European Commission, Postdoctoral Fellowship proposal (score 90%), 2025.
- **Finalist (1 of 6)**, TrustRobotics Challenge, CSAW'26 Europe, Valence, France, 2026 (team with P. Lopes and L. T. Paiva).
- Travel grants: MSCA Proposal Writing Bootcamp 2025, University of Porto; IEEE CoDIT 2023, Rome.
- **University Medal, First Rank**, M.E. Control Systems Engineering, Jadavpur University, 2017.
- Government of India national fellowship for M.E. and Ph.D. (GATE AIR 182).

Mentoring

Luis Lucas	Ph.D. student — *Deep Vision-Guided Autonomous Navigation and Control for Fixed-Wing UAVs in GNSS-Denied Environments*, Storming Universe, University of Porto (ongoing).
Tiago Vieira	M.Sc. student — *CRAZYNaV: Vision-Guided Navigation Planning in GPS-Denied Environment for Crazyflies*, University of Porto (ongoing).
Henrique Miguel Ferreira dos Santos	M.Sc. student — *Safe Overtaking Manoeuvres for Small-Scale Autonomous Racing Vehicles*, University of Porto (ongoing).
Tiago Ferreira de Araújo	B.Sc. project — *Autonomous Boundary Navigation for Surveillance Using Nano UAVs*, University of Porto.
Guilherme Reviero Pereira	B.Sc. project — *Safe Corridor Navigation in Constrained Airspace Using Nano UAV*, University of Porto.
Diogo Campos dos Santos	B.Sc. project — *Guidance and Control of Autonomous Nano UAVs for Precision Trajectory Tracking*, University of Porto.

Teaching

Mathematical Methods for Aerospace Engineers (IISc, 2021); *Guidance Theory and Applications* (IISc, 2023); industrial workshop on guidance algorithms, Bharat Electronics Ltd. (2022). Interests: guidance, navigation and control; nonlinear and robust control; estimation; autonomous aerial systems.

Talks and Service

Talks. CONTROLO, Coimbra, 2026; Airborne Wind Energy Conference, Porto, 2026; IEEE CoDIT, Rome, 2023.

Service. Technical committee member, ICAAV 2026. Reviewer for IEEE Trans. Aerospace and Electronic Systems, European Journal of Control, AIAA J. Aerospace Information Systems, IEEE ECC, ACC, CDC and ICUAS. Volunteer, IEEE CDC 2025 and AWEC 2026. IEEE member.

References available upon request.